F statistics

2024-11-11

Load

```
mad_bor_g1000_sum <-
  gt_load("../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld.gt")
```

Loci and sample count

```
nrow(mad_bor_g1000_sum)
```

```
## [1] 2744
```

```
count_loci(mad_bor_g1000_sum)
```

```
## [1] 198196
```

This script uses admixtools @maierAdmixtoolsInferringDemographic2024 to calculate F3 and F4 statistics, using the full global dataset, comprising of 2744 individuals and 4 loci.

Remove all populations with only one individual

```
pop_tbl <- table(mad_bor_g1000_sum$population)
to_remove <- which(pop_tbl == 1)
to_remove_pops <- names(to_remove)
mad_bor_g1000_sum <-
  mad_bor_g1000_sum %>% filter(!population %in% to_remove_pops)
```

Group by population

```
mad_bor_g1000_sum <- mad_bor_g1000_sum %>% group_by(population)
```

Create f2 blocks directory

```
if (dir.exists("../data/intermediate/f2")) {
  f2_dir <- "../data/intermediate/f2"
} else {
  dir.create("../data/intermediate/f2")
  gt_extract_f2(mad_bor_g1000_sum,
    outdir = "../data/intermediate/f2",
    fst = FALSE,
    n_cores = 10
  )
  f2_dir <- "../data/intermediate/f2"
}
```

Read in precomputed f2's

```
f2_blocks <- f2_from_precomp(f2_dir,
  afprod = FALSE
)
```

```
## i Reading precomputed data for 136 populations...
## i Reading f2 data for pair 1 out of 9316...i Reading f2 data for pair 2 out of 9316...i Reading f2 da
```

```
## Warning in f2_from_precomp(f2_dir, afprod = FALSE): Some f2-statistic estimates
## are negative across blocks. This is probably caused by too many missing or rare
## SNPs in populations with low sample size, which makes the f2 bias correction
## unreliable. Consider running 'f2_from_precomp' with 'afprod = TRUE'.
```

```
dim(f2_blocks)
```

```
## [1]   136   136 45175
```

```
count_snps(f2_blocks)
```

```
## [1] 197972
```

Test for admixture: F3(Malagasy; Asian, African)

```
## Running C = GWD
```

```
## Running C = ESN
```

```
## Running C = MSL
```

```
## Running C = YRI
```

```
## Running C = LWK
```

```
## Running C = Botswana_Kalanga

## Running C = Zimbabwe_Remba

## Running C = CAR_Gbaya

## Running C = CAR_DzangaShangaPeople

## Running C = CAR_Mpiemo

## Running C = CAR_Banda

## Running C = CAR_Nzakara

## Running C = CAR_Zande

## Running C = SouthAfrica_ColouredWellington

## Running C = SouthAfrica_Swazi

## Running C = SouthAfrica_Karretjie

## Running C = SouthAfrica_ColouredColesberg

## Running C = SouthAfrica_Xhosa

## Running C = Namibia_Damara

## Running C = Namibia_Himba

## Running C = Angola_Khwe

## Running C = Botswana_Kgalagadi

## Running C = SouthAfrica_Khomani

## Running C = SouthAfrica_SothoSouthern

## Running C = SouthAfrica_Zulu

## Running C = SouthAfrica_Bhaca

## Running C = SouthAfrica_Hlubi

## Running C = SouthAfrica_Mputhi

## Running C = Mozambique_Nhungwe
```

```
## Running C = Mozambique_Tsonga

## Running C = Mozambique_Ronga

## Running C = Mozambique_TongaMoz

## Running C = Mozambique_Chuwabu

## Running C = Mozambique_Tswa

## Running C = Mozambique_Chopi

## Running C = SouthAfrica_Ndebele

## Running C = Angola_Xun

## Running C = Namibia_Wambo

## Running C = SouthAfrica_SothoNorthern

## Running C = Botswana_TswanaBot

## Running C = Botswana_GuiGhanaKgal

## Running C = Lesotho_SothoSouthern

## Running C = Swaziland_Swazi

## Running C = DRC_Manyanga

## Running C = DRC_Yombe

## Running C = DRC_Ndibu

## Running C = DRC_MongoNkundo

## Running C = DRC_LubaLulua

## Running C = DRC_Ntandu

## Running C = DRC_Shi

## Running C = DRC_Tetela

## Running C = DRC_Yans

## Running C = DRC_Mbala
```

```
## Running C = DRC_Nandi

## Running C = DRC_Yaka

## Running C = DRC_Ngombe

## Running C = DRC_Pende

## Running C = DRC_Songe

## Running C = DRC_Kuba

## Running C = DRC_Lunda

## Running C = SouthAfrica_Tsonga

## Running C = SouthAfrica_Tswana

## Running C = SouthAfrica_Venda

## Running C = Rwanda_Nkore

## Running C = Uganda_Konzo

## Running C = Uganda_Kiga

## Running C = Uganda_Fumbira

## Running C = Uganda_Nkore

## Running C = Zambia_Chewa

## Running C = Zambia_Kaonde

## Running C = Zambia_Lozi

## Running C = Zambia_ShonaZam

## Running C = Zambia_Bemba

## Running C = Zambia_Tumbuka

## Running C = Zambia_TongaZam

## Running C = Tanzania_TanzaniaMixed

## Running C = Zanzibar_Swahili
```

```
## Running C = Zambia_Ngoni

## Running C = Zambia_Nsenga

## Running C = SouthAfrica_Coloured

## Running C = Angola_Vasekele

## Running C = Namibia_TsumkweKung

## Running C = Namibia_Herero

## Running C = Namibia_Nama

## Running C = Angola_Sekele

## Running C = Namibia_Kgalagadi

## Running C = DRC_Luozi

## Running C = Zambia_Tswana

## Running C = Zambia_Kalanga

## Running C = Zambia_Kgalagadi

## Running C = Zambia_Mbukushu

## Running C = Zambia_Mbunda

## Running C = Zambia_Kwangwa

## Running C = Zambia_Luyana

## Running C = Zambia_Nkoya

## Running C = Zambia_Nyengo

## Running C = Zambia_Shanjo

## Running C = Zambia_Kwamashi

## Running C = Zambia_Fwe

## Running C = DRC_Ding

## Running C = DRC_Wonk
```

```
## Running C = DRC_Lwer

## Running C = DRC_Nsong

## Running C = DRC_Mbuun

## Running C = DRC_Nzadi

## Running C = DRC_Hungan

## Running C = DRC_Lele

## Running C = DRC_Mpiin

## Running C = DRC_Ngwi

## Running C = DRC_LubaKasai

## Running C = DRC_Ngong

## Running C = DRC_Sakata

## Running C = DRC_Mpe

## Running C = DRC_NunuBobangi

## Running C = DRC_Mbudza

## Running C = DRC_LubaKatanga

## Running C = DRC_Rega

## Running C = DRC_BanguBangu

## Running C = SouthAfrica_Pedi

## Running C = SouthAfrica_Sotho

## Running C = Botswana_Ghanzi
```

```r
test <- do.call(rbind, results)

colnames(test) <- c("Malagasy", "Asian", "African", "est", "se", "z", "p")

test$Asian <- sub("BOR", "Borneo", test$Asian)
test$Asian <- sub("CHS", "Han, China", test$Asian)
test$Asian <- sub("KHV", "Kinh, Vietnam", test$Asian)
test$Asian <- sub("PJL", "Punjabi, Pakistan", test$Asian)
```

```r
test$Asian <- sub("BEB", "Bengali, Bengal", test$Asian)

test$African <- sub("ESN", "Esan, Nigeria", test$African)
test$African <- sub("YRI", "Yoruba, Nigeria", test$African)
test$African <- sub("GWD", "Mandinka, The Gambia", test$African)
test$African <- sub("LWK", "Luhye, Kenya", test$African)
test$African <- sub("MSL", "Mende, Sierra Leone", test$African)

test$Malagasy <- sapply(
  strsplit(test$Malagasy, "_"),
  function(x) paste(x, collapse = " ")
)

print(test)
```

```
## # A tibble: 6,776 x 7
##    Malagasy      Asian             African              est      se      z     p
##    <chr>         <chr>             <chr>              <dbl>   <dbl>  <dbl> <dbl>
##  1 Betsileo      Bengali, Bengal   Mandinka, The ~ -0.00369 5.69e-5  -64.8     0
##  2 Betsileo      Borneo            Mandinka, The ~ -0.0110  5.65e-5 -195.      0
##  3 Betsileo      Han, China        Mandinka, The ~ -0.0106  5.86e-5 -180.      0
##  4 Betsileo      Kinh, Vietnam     Mandinka, The ~ -0.0107  5.68e-5 -189.      0
##  5 Betsileo      Punjabi, Pakistan Mandinka, The ~ -0.00243 6.01e-5  -40.4     0
##  6 Betsileo      Sumba             Mandinka, The ~ -0.00940 5.87e-5 -160.      0
##  7 Betsileo      Timor             Mandinka, The ~ -0.00815 7.21e-5 -113.      0
##  8 Betsimisaraka Bengali, Bengal   Mandinka, The ~ -0.00546 4.92e-5 -111.      0
##  9 Betsimisaraka Borneo            Mandinka, The ~ -0.0104  5.40e-5 -192.      0
## 10 Betsimisaraka Han, China        Mandinka, The ~ -0.0101  5.52e-5 -182.      0
## # i 6,766 more rows
```

Admixture f3 statistics were used in the configuration f3(Malagasy; Asian, African) to examine which comparative African and Asian populations show the greatest similarity to the 'source' populations of the Malagasy.

Examine min F3 values for each Malagasy popualtion

```r
obj <- test %>%
  group_by(Malagasy) %>%
  slice(which.min(est))
obj
```

```
## # A tibble: 8 x 7
## # Groups:   Malagasy [8]
##   Malagasy           Asian  African              est        se      z     p
##   <chr>              <chr>  <chr>              <dbl>     <dbl>  <dbl> <dbl>
## 1 Betsileo           Borneo Mozambique_Ronga -0.0124 0.0000925 -134.      0
## 2 Betsimisaraka      Borneo Mozambique_Chopi -0.0120 0.0000810 -149.      0
## 3 Diego              Borneo Mozambique_Tswa  -0.0113 0.0000828 -137.      0
## 4 Merina             Borneo Mozambique_Tswa  -0.0112 0.0000747 -149.      0
## 5 Mikea              Borneo Mozambique_Ronga -0.0105 0.000111   -94.7     0
## 6 Sakalava Tsimihety Borneo Mozambique_Chopi -0.0122 0.0000811 -150.      0
```

```
## 7 Temoro             Borneo Mozambique_Ronga -0.0120 0.000104  -115.      0
## 8 Vezo               Borneo Mozambique_Chopi -0.0120 0.0000829 -145.      0
```

```
write.table(
  file = paste0(res_dir, "minimum_f3_malagasy_pops"),
  obj, row.names = FALSE, quote = FALSE, sep = "\t"
)
```

```
# Examine z-scores remembering that z-score is affected by population size
min_z <- test %>%
  group_by(Malagasy) %>%
  slice(which.min(z))
min_z
```

```
## # A tibble: 8 x 7
## # Groups:   Malagasy [8]
##   Malagasy           Asian  African       est        se     z     p
##   <chr>              <chr>  <chr>       <dbl>     <dbl> <dbl> <dbl>
## 1 Betsileo           Borneo DRC_Shi  -0.0115  0.0000551 -209.     0
## 2 Betsimisaraka      Borneo DRC_Shi  -0.0110  0.0000516 -213.     0
## 3 Diego              Borneo DRC_Shi  -0.0102  0.0000485 -211.     0
## 4 Merina             Borneo DRC_Shi  -0.0105  0.0000567 -185.     0
## 5 Mikea              Borneo DRC_Ngwi -0.00991 0.0000614 -161.     0
## 6 Sakalava Tsimihety Borneo DRC_Shi  -0.0112  0.0000508 -220.     0
## 7 Temoro             Borneo DRC_Shi  -0.0110  0.0000537 -205.     0
## 8 Vezo               Borneo DRC_Shi  -0.0110  0.0000546 -201.     0
```

```
mad_bor_g1000_sum %>%
  filter(population == "DRC_Ngwi") %>%
  nrow()
```

```
## [1] 67
```

```
mad_bor_g1000_sum %>%
  filter(population == "DRC_Shi") %>%
  nrow()
```

```
## [1] 129
```

```
mad_bor_g1000_sum %>%
  filter(population == "Mozambique_Ronga") %>%
  nrow()
```

```
## [1] 2
```

```
mad_bor_g1000_sum %>%
  filter(population == "Mozambique_Chopi") %>%
  nrow()
```

```
## [1] 4
```

```r
mad_bor_g1000_sum %>%
  filter(population == "Mozambique_Tswa") %>%
  nrow()
```

```
## [1] 4
```

```r
write.table(
  file = paste0(res_dir, "minimum_z-score_malagasy_pops"),
  min_z, row.names = FALSE, quote = FALSE, sep = "\t"
)
```

F3 values for each Malagasy population descending all combinations

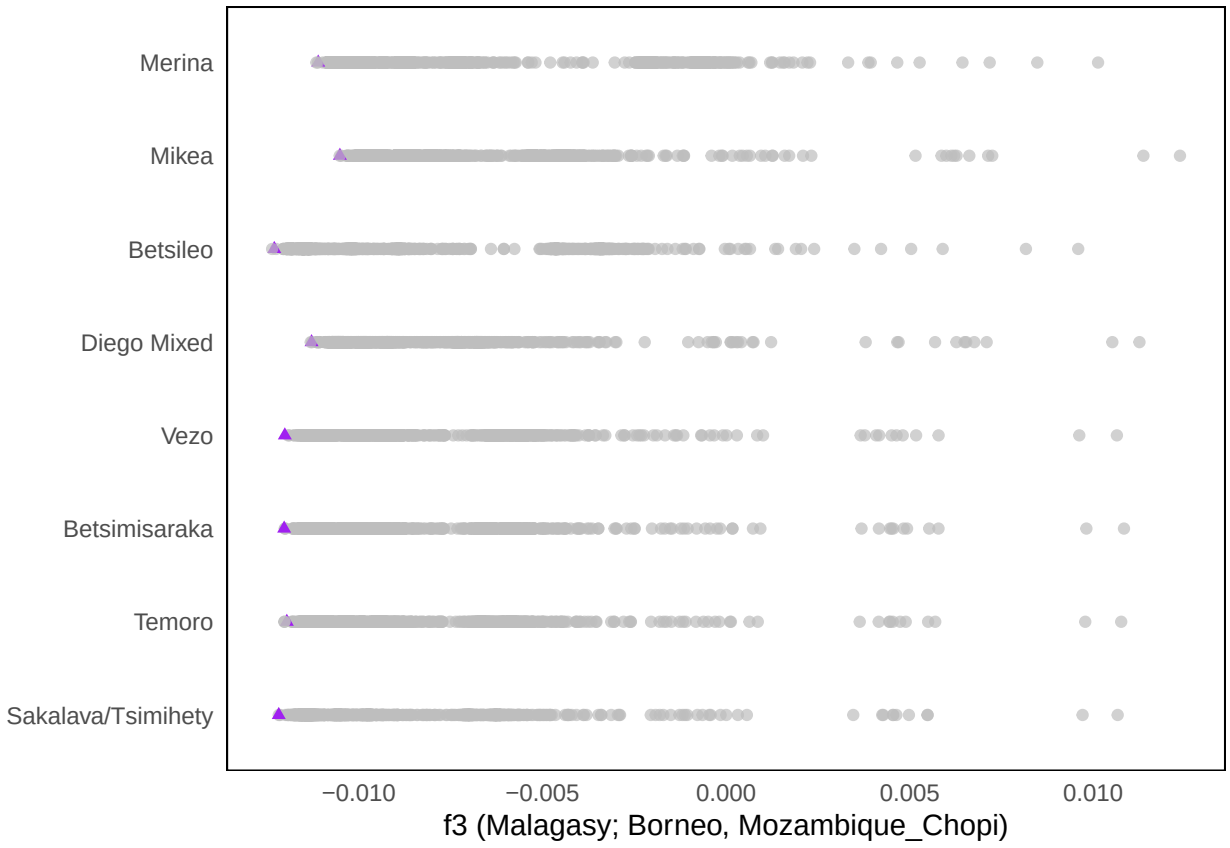
```r
# update labels
test <- test %>% mutate(Malagasy = case_when(
  Malagasy == "Sakalava Tsimihety" ~ "Sakalava/Tsimihety",
  TRUE ~ Malagasy
))
test <- test %>% mutate(Malagasy = case_when(
  Malagasy == "Diego" ~ "Diego Mixed",
  TRUE ~ Malagasy
))

test_grouped <- test %>%
  select(est, se, z, Malagasy, Asian, African) %>%
  reframe(est = est, se = se, z, Malagasy, Asian, African)

test_grouped$highlight <- "NA"
# find rows with"Mozambique_Chopi" and"Borneo" as references
test_grouped$highlight[which(test_grouped$African == "Mozambique_Chopi" & test_grouped$Asian == "Borneo
test_grouped$highlight <- as.factor(test_grouped$highlight)

test_grouped %>%
  group_by(Malagasy) %>%
  arrange(desc(est)) %>%
  ggplot(aes(
    x = est, y = reorder(Malagasy, est),
    color = highlight, shape = highlight, alpha = highlight
  )) +
  geom_point() +
  labs(
    x = "f3 (Malagasy; Borneo, Mozambique_Chopi)", y = NULL,
    color = element_blank(), shape = element_blank(),
    alpha = element_blank()
  ) +
  scale_shape_manual(
    values = c(
      "NA" = 19, # "Borneo" = 3, "Mozambique_Chopi" = 15,
      "Borneo and Mozambique_Chopi" = 17
```

```
  ),
  breaks = c(
    # "Borneo", "Mozambique_Chopi",
    "Borneo and Mozambique_Chopi"
  )
) +
scale_color_manual(
  values = c(
    "NA" = "grey", # "Borneo" = "lightpink", "Mozambique_Chopi" = "lightblue",
    "Borneo and Mozambique_Chopi" = "purple"
  ),
  breaks = c(
    # "Borneo", "Mozambique_Chopi",
    "Borneo and Mozambique_Chopi"
  )
) +
scale_alpha_manual(
  values = c(
    "NA" = 0.7, # "Borneo" = 0.7, "Mozambique_Chopi" = 0.7,
    "Borneo and Mozambique_Chopi" = 1
  ),
  breaks = c(
    # "Borneo", "Mozambique_Chopi",
    "Borneo and Mozambique_Chopi"
  )
) +
theme_minimal() +
theme(
  legend.position = "none",
  axis.line = element_blank(),
  panel.grid.major = element_blank(),
  panel.grid.minor = element_blank(),
  panel.background = element_rect(fill = "white")
)
```




```
ggsave(
  filename = paste0(res_dir, "F3_all_combinations.jpg"),
  width = 7, height = 5, units = "in", dpi = 300
)
```

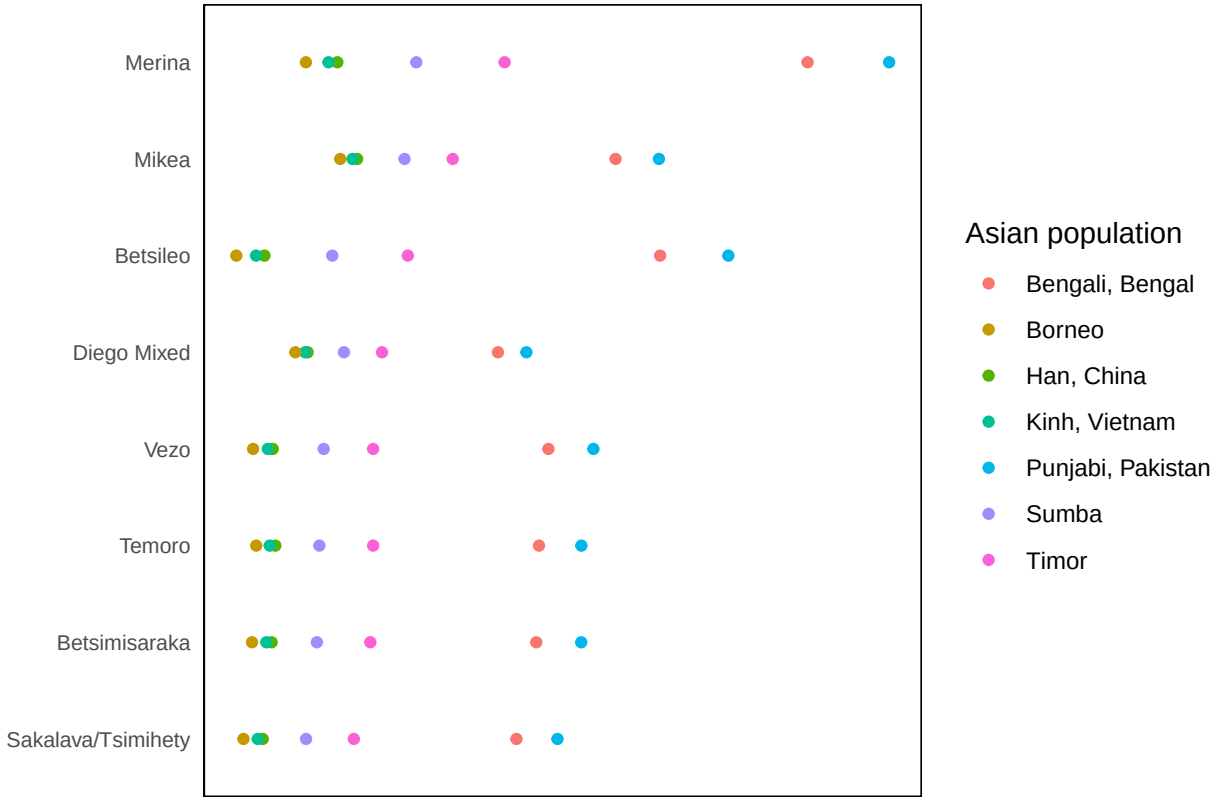
Examining f3 for each Asian reference population

F3 values for each Malagasy population grouped, using Mozambique_Chopi as the African reference.

```
adm_f3 <- test %>%
  filter(African == "Mozambique_Chopi") %>%
  select(est, se, z, Malagasy, Asian) %>%
  reframe(est = est, se = se, z, Malagasy, Asian)


adm_f3 %>%
  group_by(Malagasy) %>%
  arrange(desc(est)) %>%
  ggplot(aes(x = est, y = reorder(Malagasy, est), colour = Asian)) +
  geom_point() +
  labs(
    x = "f3 (Malagasy; Asian reference, Mozambique_Chopi)", y = NULL,
    colour = "Asian population"
  ) +
```

```
  theme_minimal() +
  theme(
    axis.text.x = element_blank(),
    axis.text.y = element_text(size = 8),
    legend.position = "right",
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white")
  )
```



f3 (Malagasy; Asian reference, Mozambique_Chopi)

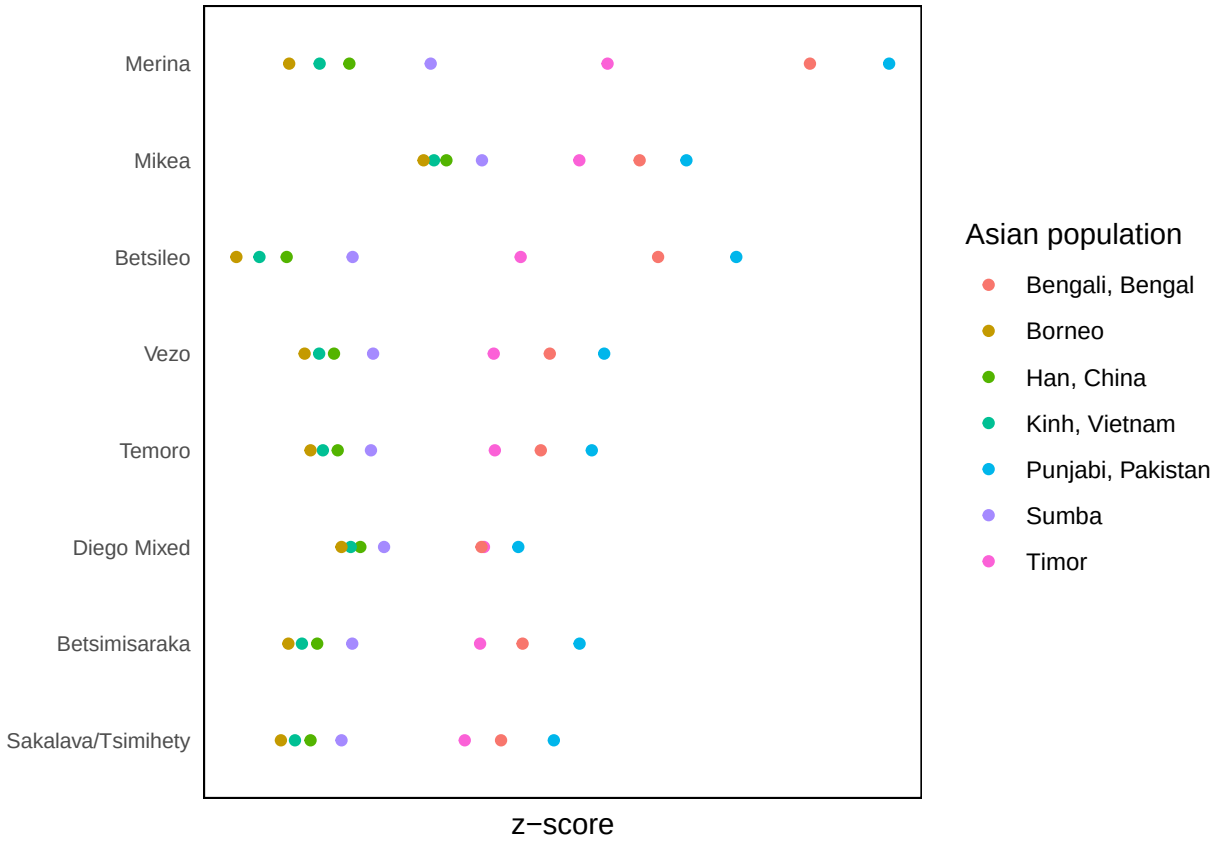
```
ggsave(
  filename = paste0(res_dir, "F3_Asian_source_per_malagasy_pop.jpg"),
  width = 7, height = 5, units = "in", dpi = 300
)
```

Also check z-scores below

```
adm_f3 <- test %>%
  filter(African == "Mozambique_Chopi") %>%
  select(est, se, z, Malagasy, Asian) %>%
  reframe(est = est, se = se, z, Malagasy, Asian)


adm_f3 %>%
```

```
  group_by(Malagasy) %>%
  arrange(desc(z)) %>%
  ggplot(aes(x = z, y = reorder(Malagasy, z), colour = Asian)) +
  geom_point() +
  labs(
    x = "z-score", y = NULL,
    colour = "Asian population"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_blank(),
    axis.text.y = element_text(size = 8),
    legend.position = "right",
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white")
  )
```




```
ggsave(
  filename = paste0(res_dir, "F3_Asian_source_per_malagasy_pop_z.jpg"),
  width = 7, height = 5, units = "in", dpi = 300
)
```

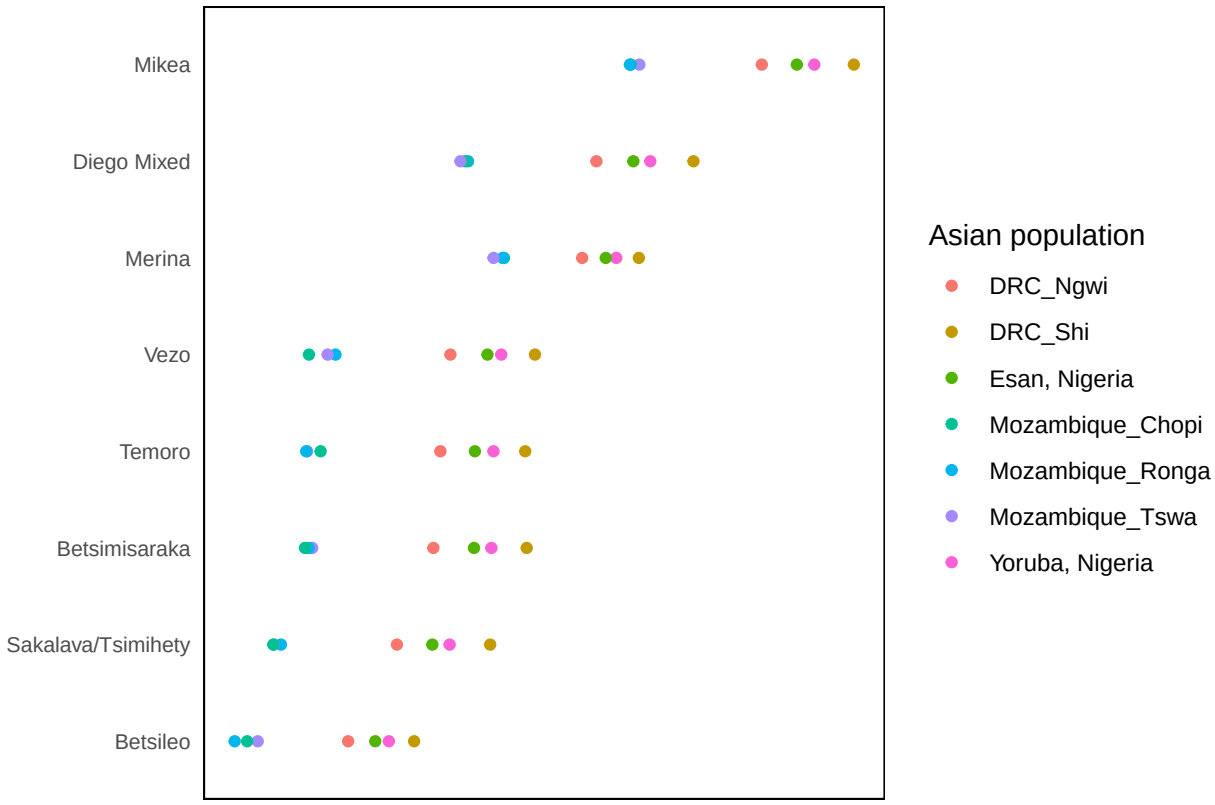
Examining f3 for each African reference population

There are many African popualtions in the dataset, here we will focus on the ones with the lowest f3 values, which are likely to be the most similar to the African source population for the Malagasy.

```
adm_f3 <- test %>%
  filter(Asian == "Borneo") %>%
  filter(African %in% c("Mozambique_Chopi", "Mozambique_Ronga", "Mozambique_Tswa",
                        "DRC_Shi", "DRC_Ngwi", "Yoruba, Nigeria", "Esan, Nigeria")) %>%
  reframe(est = est, se = se, z, Malagasy, African)

plt <- adm_f3 %>%
  group_by(Malagasy) %>%
  arrange(desc(est)) %>%
  ggplot(aes(x = est, y = reorder(Malagasy, est), colour = African)) +
  geom_point() +
  labs(
    x = "f3 (Malagasy; Borneo, African reference)", y = NULL,
    colour = "Asian population"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_blank(),
    axis.text.y = element_text(size = 8),
    legend.position = "right",
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white")
  )

plt
```



f3 (Malagasy; Borneo, African reference)

```
ggsave(
  filename = paste0(res_dir, "F3_African_source_per_malagasy_pop.jpg"),
  width = 7, height = 5, units = "in", dpi = 300
)
```

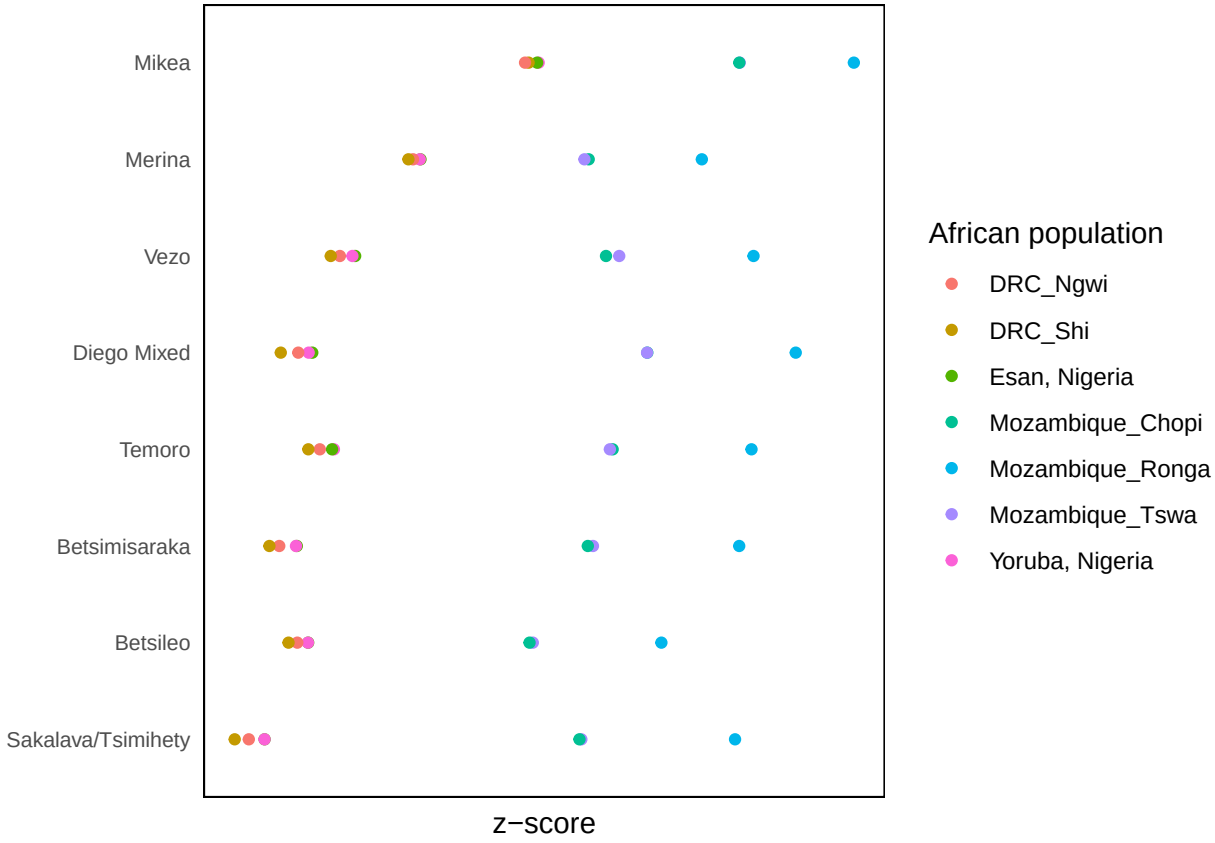
Also check z-scores

```
adm_f3 <- test %>%
  filter(Asian == "Borneo") %>%
  filter(African %in% c("Mozambique_Chopi", "Mozambique_Ronga", "Mozambique_Tswa",
                        "DRC_Shi", "DRC_Ngwi", "Yoruba, Nigeria", "Esan, Nigeria")) %>%
  select(est, se, z, Malagasy, African) %>%
  reframe(est = est, se = se, z, Malagasy, African)


plt <- adm_f3 %>%
  group_by(Malagasy) %>%
  arrange(desc(z)) %>%
  ggplot(aes(x = z, y = reorder(Malagasy, z), colour = African)) +
  geom_point() +
  labs(x = "z-score", y = NULL,
    colour = "African population"
  ) +
  theme_minimal() +
  theme(
```

```
    axis.text.x = element_blank(),
    axis.text.y = element_text(size = 8),
    legend.position = "right",
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white")
  )

plt
```




```
ggsave(
  filename = paste0(res_dir, "F3_African_source_per_malagasy_pop_z.jpg"),
  width = 7, height = 5, units = "in", dpi = 300
)
```

Minimum z-scores were observed using the DRC_Shi or DRC_Ngwi populations (n = 129 and n = 67 respectively), likely due to their much larger sample size in comparison to the Mozambique populations (Mozambique_Ronga, n = 2; Mozambique_Chopi, n = 4; Mozambique_Tswa, n = 4).

F4 ratio

To estimate the proportion of each ancestry source within the Malagasy population, we will use an f4 ratio in the following configurations:

f4(GBR,Gorilla;Malagasy,Borneo)/f4(GBR,Chimpanzee;Esan,Borneo)

f4(GBR,Gorilla;Malagasy,Esan)/f4(GBR,Chimpanzee;Borneo,Esan)

Load

```
mad_bor_g1000_sum <-
  gt_load("../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld.gt")
```

Loci and sample count

```
nrow(mad_bor_g1000_sum)
```

```
## [1] 2744
```

```
count_loci(mad_bor_g1000_sum)
```

```
## [1] 198196
```

Split Merina

```
qmats <- read_q_files("../data/intermediate/admixture")

fam <- read.table(
  "../data/intermediate/admixture/admixture_supervised_subset.fam",
  header = FALSE
)

qmats$id <- fam$V2

# subset mad_bor_g1000_sum to only individuals in qmats$id
mad_bor_g1000_sum_sub <- mad_bor_g1000_sum %>%
  filter(id %in% qmats$id)

# tidy with the gt
mad_bor_g1000_sum_sub <- augment(get_q_matrix(qmats, k = 2, run = 1),
  data = mad_bor_g1000_sum_sub
)
# filter to merina
merina <- mad_bor_g1000_sum_sub %>% filter(population == "Merina")

# save output
saveRDS(merina, file = paste0(res_dir, "merina_qmats.rds"))

# split groups
```

```
# .Q3 representing Asian ancestry and .Q2 representing African ancestry
merina$population <- ifelse(merina$.Q1 > 0.5, "MerinaA", "MerinaB")
merina <- merina[, c("id", "population")]

# match V2 and merina$id and replace the population column in merina_fam with
# the population column in merina
matching_m <- match(mad_bor_g1000_sum$id, merina$id)
mad_bor_g1000_sum$population <- ifelse(!is.na(matching_m),
  merina$population[matching_m], mad_bor_g1000_sum$population
)

# count number of MerinaA
cat(
  "Merina group > 0.5",
  length(which(mad_bor_g1000_sum$population == "MerinaA"))
)
```

```
## Merina group > 0.5 23
```

```
cat(
  "Merina group < 0.5",
  length(which(mad_bor_g1000_sum$population == "MerinaB"))
)
```

```
## Merina group < 0.5 11
```

Remove all populations with only one individual

```
pop_tbl <- table(mad_bor_g1000_sum$population)
to_remove <- which(pop_tbl == 1)
to_remove_pops <- names(to_remove)
mad_bor_g1000_sum <-
  mad_bor_g1000_sum %>% filter(!population %in% to_remove_pops)
```

Group by population

```
mad_bor_g1000_sum <- mad_bor_g1000_sum %>% group_by(population)
```

Add Chimpanzee outgroup for f4

```
mad_bor_g1000_sum <- mad_bor_g1000_sum %>% ungroup()
chimp <- gt_load("../data/intermediate/pan_tro_liftover/chimp_ref_v2.gt")
f4_merged_data <- rbind(mad_bor_g1000_sum, chimp,
  as_is = FALSE, flip_strand = TRUE,
```

```
  quiet = FALSE, use_position = FALSE,
  backingfile = tempfile()
)
```

```
## harmonising loci between two datasets
## flip_strand =  TRUE  ; remove_ambiguous =  TRUE
## ------------------------------
## dataset: reference
## number of SNPs: 198196 reduced to 77993
## ( 0 are ambiguous, of which 0  were removed)
## ------------------------------
## dataset: target
## number of SNPs: 77993 reduced to 77993
## ( 0 were flipped to match the reference set)
## ( 0 are ambiguous, of which 0 were removed)

##
## gen_tibble saved to /tmp/RtmpHqh5nN/file621c4498a1c7.gt

## using FBM RDS: /tmp/RtmpHqh5nN/file621c4498a1c7.rds

## with FBM backing file: /tmp/RtmpHqh5nN/file621c4498a1c7.bk

## make sure that you do NOT delete those files!

## to reload the gen_tibble in another session, use:

## gt_load('/tmp/RtmpHqh5nN/file621c4498a1c7.gt')
```

```
g1000 <- gt_load("../data/raw/data_input/1000_Genomes_b37_clean.gt")
# append population info
pop_table <- read.table("../data/raw/data_input/Madagascar_Borneo.poptable",
  sep = "\t", fill = TRUE
)
colnames(pop_table) <- c("IID", "V2", "FID", "Population", "V5", "V6", "V7")

pop_table_sub <- subset(pop_table, pop_table$IID %in% g1000$id)

matching_m <- match(g1000$id, pop_table_sub$IID)
g1000$population <- ifelse(!is.na(matching_m),
  pop_table_sub$V2[matching_m],
  g1000$population
)

british <- g1000 %>% filter(population == "GBR")

show_loci(british)$name <- paste(show_loci(british)$chromosome,
  show_loci(british)$position,
  sep = "_"
)
```

```
f4_merged_data <- rbind(f4_merged_data, british,
  as_is = FALSE, flip_strand = TRUE,
  quiet = FALSE, use_position = FALSE,
  backingfile = tempfile()
)
```

```
## harmonising loci between two datasets
## flip_strand =  TRUE  ; remove_ambiguous =  TRUE
## ------------------------------
## dataset: reference
## number of SNPs: 77993 reduced to 77993
## ( 0 are ambiguous, of which 0  were removed)
## ------------------------------
## dataset: target
## number of SNPs: 670552 reduced to 77993
## ( 1777 were flipped to match the reference set)
## ( 3 are ambiguous, of which 3 were removed)

##
## gen_tibble saved to /tmp/RtmpHqh5nN/file621c21478b38.gt

## using FBM RDS: /tmp/RtmpHqh5nN/file621c21478b38.rds

## with FBM backing file: /tmp/RtmpHqh5nN/file621c21478b38.bk

## make sure that you do NOT delete those files!

## to reload the gen_tibble in another session, use:

## gt_load('/tmp/RtmpHqh5nN/file621c21478b38.gt')
```

```
nrow(f4_merged_data)
```

```
## [1] 2792
```

```
f4_merged_data <- f4_merged_data %>% group_by(population)
```

Create f2 blocks directory

```
if (dir.exists("../data/intermediate/f2_outgroup")) {
  f2_dir <- "../data/intermediate/f2_outgroup"
} else {
  dir.create("../data/intermediate/f2_outgroup")
  gt_extract_f2(f4_merged_data,
    outdir = "../data/intermediate/f2_outgroup"
  )
}
```

Read in precomputed f2's

```
f2_dir <- "../data/intermediate/f2_outgroup"
f2_blocks <- f2_from_precomp(f2_dir, afprod = TRUE)
```

```
## i Reading precomputed data for 139 populations...
## i Reading ap data for pair 1 out of 9730...i Reading ap data for pair 2 out of 9730...i Reading ap da
```

```
dim(f2_blocks)
```

```
## [1]   139   139 24966
```

```
count_snps(f2_blocks)
```

```
## [1] 77911
```

F4 ratio: estimating the proportion of African admixture

Sister taxon and outgroup on the left, target in the centre, and sister population and source of admixture on the right.

```
malagasy <- c(
  "Betsimisaraka", "MerinaA", "MerinaB", "Diego", "Sakalava_Tsimihety",
  "Betsileo", "Temoro", "Vezo", "Mikea"
)

pop_matrix <-
  as.matrix(data.frame("GBR", "chimp_ref", malagasy, "Borneo", "ESN"))

qp4_african_p <- qpf4ratio(f2_blocks, pop_matrix)

qp4_african_p$pop3 <- factor(qp4_african_p$pop3,
  levels = c(
    "Betsimisaraka", "MerinaA", "MerinaB", "Diego",
    "Sakalava_Tsimihety", "Betsileo",
    "Temoro", "Vezo", "Mikea"
  )
)
# order by qp4_african_p$pop3 levels
qp4_african_p <- qp4_african_p %>% mutate(pop3 = forcats::fct_inorder(pop3))
# round alpha to 1 decimal place
qp4_african_p
```

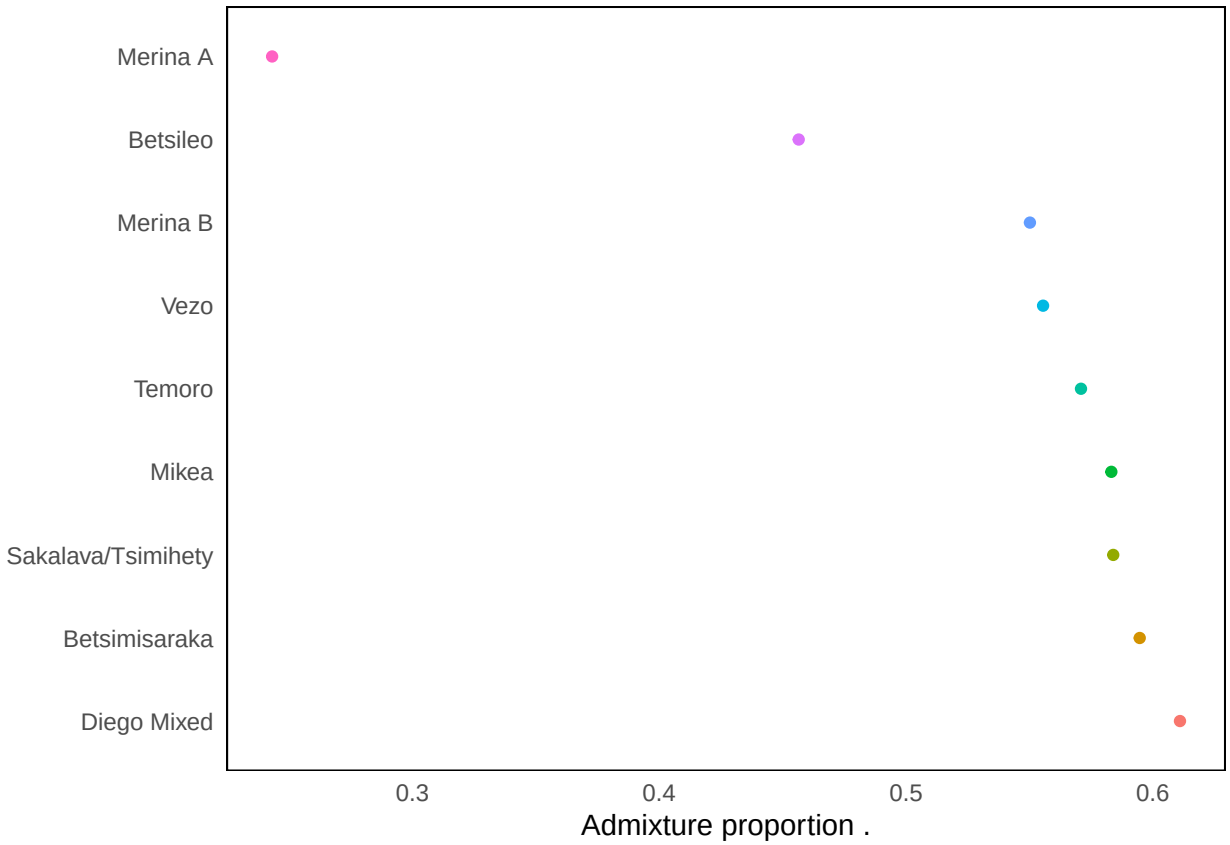
```
## # A tibble: 9 x 8
##   pop1  pop2      pop3              pop4   pop5  alpha      se     z
##   <chr> <chr>     <fct>             <chr>  <chr> <dbl>   <dbl> <dbl>
## 1 GBR   chimp_ref Betsimisaraka     Borneo ESN   0.595 0.00903  65.9
## 2 GBR   chimp_ref MerinaA           Borneo ESN   0.243 0.0120   20.2
```

```
## 3 GBR   chimp_ref MerinaB            Borneo ESN   0.550 0.0134   41.1
## 4 GBR   chimp_ref Diego              Borneo ESN   0.611 0.00906  67.4
## 5 GBR   chimp_ref Sakalava_Tsimihety Borneo ESN   0.584 0.00893  65.4
## 6 GBR   chimp_ref Betsileo           Borneo ESN   0.456 0.00955  47.8
## 7 GBR   chimp_ref Temoro             Borneo ESN   0.571 0.00990  57.7
## 8 GBR   chimp_ref Vezo               Borneo ESN   0.555 0.0102   54.7
## 9 GBR   chimp_ref Mikea              Borneo ESN   0.583 0.0116   50.4
```

```r
# save output
write.table(
  file = paste0(res_dir, "f4_ratio_african_admixture"),
  qp4_african_p, row.names = FALSE, quote = FALSE, sep = "\t"
)
```

```r
# update labels
# rename qp4_african_p$pop3 "Sakalava Tsimihety" to "Sakalava/Tsimihety"
qp4_african_p <- qp4_african_p %>% mutate(pop3 = case_when(
  pop3 == "Sakalava_Tsimihety" ~ "Sakalava/Tsimihety",
  TRUE ~ pop3
))
# rename qp4_african_p$pop3 "Diego" to "Diego Mixed"
qp4_african_p <- qp4_african_p %>% mutate(pop3 = case_when(
  pop3 == "Diego" ~ "Diego Mixed",
  TRUE ~ pop3
))
# rename qp4_african_p$pop3 "MerinaA" to "Merina A"
qp4_african_p <- qp4_african_p %>% mutate(pop3 = case_when(
  pop3 == "MerinaA" ~ "Merina A",
  TRUE ~ pop3
))
# rename qp4_african_p$pop3 "MerinaB" to "Merina B"
qp4_african_p <- qp4_african_p %>% mutate(pop3 = case_when(
  pop3 == "MerinaB" ~ "Merina B",
  TRUE ~ pop3
))

qp4_african_p <- qp4_african_p %>%
  arrange(desc(alpha)) %>%
  mutate(pop3 = forcats::fct_inorder(pop3))
ggplot(qp4_african_p, aes(x = alpha, y = pop3, colour = pop3, fill = pop3)) +
  geom_point() +
  labs(
    x = paste("Admixture proportion \u03b1"), y = NULL,
    colour = "Malagasy population", fill = "Malagasy population"
  ) +
  guides(colour = "none", fill = "none") +
  theme_minimal() +
  theme(
    legend.position = "right",
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white")
  )
```




```
ggsave(
  filename = paste0(res_dir, "F4_African_source.jpg"),
  width = 7, height = 5, units = "in", dpi = 300
)
```

Merina is lowest, as expected.

```
pop_matrix <-
  as.matrix(data.frame("GBR", "chimp_ref", malagasy, "ESN", "Borneo"))

qp4_asian_p <- qpf4ratio(f2_blocks, pop_matrix)

qp4_asian_p$pop3 <- factor(
  qp4_asian_p$pop3,
  c(
    "Betsimisaraka", "Merina", "Diego",
    "Sakalava_Tsimihety", "Betsileo", "Temoro",
    "Vezo", "Mikea"
  )
)

qp4_asian_p <- qp4_asian_p %>% mutate(pop3 = forcats::fct_inorder(pop3))
qp4_asian_p
```

```
## # A tibble: 9 x 8
```

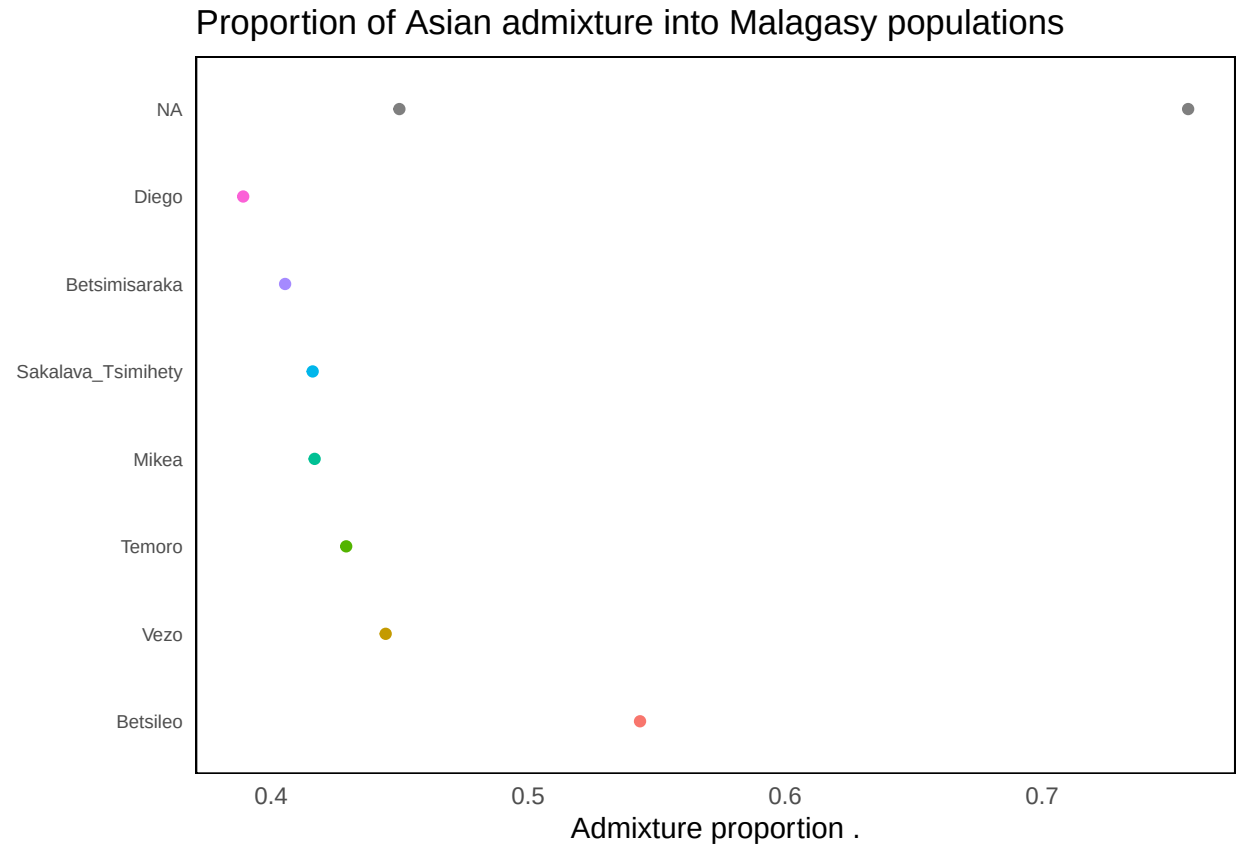
```
##   pop1  pop2      pop3                pop4  pop5   alpha      se     z
##   <chr> <chr>     <fct>               <chr> <chr>  <dbl>   <dbl> <dbl>
## 1 GBR   chimp_ref Betsimisaraka       ESN   Borneo 0.405 0.00903  44.9
## 2 GBR   chimp_ref <NA>                ESN   Borneo 0.757 0.0120   62.9
## 3 GBR   chimp_ref <NA>                ESN   Borneo 0.450 0.0134   33.6
## 4 GBR   chimp_ref Diego               ESN   Borneo 0.389 0.00906  42.9
## 5 GBR   chimp_ref Sakalava_Tsimihety ESN   Borneo 0.416 0.00893  46.6
## 6 GBR   chimp_ref Betsileo            ESN   Borneo 0.544 0.00955  56.9
## 7 GBR   chimp_ref Temoro              ESN   Borneo 0.429 0.00990  43.3
## 8 GBR   chimp_ref Vezo                ESN   Borneo 0.445 0.0102   43.8
## 9 GBR   chimp_ref Mikea               ESN   Borneo 0.417 0.0116   36.0
```

Merina and Betsileo are highest, as expected.

```
# update labels
# rename qp4_african_p$pop3 "Sakalava Tsimihety" to "Sakalava/Tsimihety"
qp4_african_p <- qp4_african_p %>% mutate(pop3 = case_when(
  pop3 == "Sakalava_Tsimihety" ~ "Sakalava/Tsimihety",
  TRUE ~ pop3
))
# rename qp4_african_p$pop3 "Diego" to "Diego Mixed"
qp4_african_p <- qp4_african_p %>% mutate(pop3 = case_when(
  pop3 == "Diego" ~ "Diego Mixed",
  TRUE ~ pop3
))
# rename qp4_african_p$pop3 "MerinaA" to "Merina A"
qp4_african_p <- qp4_african_p %>% mutate(pop3 = case_when(
  pop3 == "MerinaA" ~ "Merina A",
  TRUE ~ pop3
))
# rename qp4_african_p$pop3 "MerinaB" to "Merina B"
qp4_african_p <- qp4_african_p %>% mutate(pop3 = case_when(
  pop3 == "MerinaB" ~ "Merina B",
  TRUE ~ pop3
))

qp4_asian_p <- qp4_asian_p %>%
  arrange(desc(alpha)) %>%
  mutate(pop3 = forcats::fct_inorder(pop3))
ggplot(qp4_asian_p, aes(x = alpha, y = pop3, colour = pop3, fill = pop3)) +
  geom_point() +
  labs(
    title = "Proportion of Asian admixture into Malagasy populations",
    x = paste("Admixture proportion \u03b1"), y = NULL,
    colour = "Malagasy population", fill = "Malagasy population"
  ) +
  guides(colour = "none", fill = "none") +
  theme_minimal() +
  theme(
    # axis.text.x = element_blank(),
    axis.text.y = element_text(size = 7),
    legend.position = "right",
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
```

```
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white")
  )
```




```
ggsave(
  filename = paste0(res_dir, "F4_Asian_source.jpg"),
  width = 7, height = 5, units = "in", dpi = 300
)
```